SERIES: A

QP CODE: 24900084

SERIES: A **Reg No:.....**

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Multi-Disciplinary Course - MG1MDCECT101 - FOUNDATION OF AI AUTOMATION

(2024 ADMISSION ONWARDS)

Duration: 1 Hours Maximum Marks: 35

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A

Multiple Choice Questions

Answer any 35 questions.Each question carries one mark

1 What is Narrow AI primarily designed to do? [U] [1]

a) General problem solving b) Specific tasks

c) Human-like reasoning d) Creative thinking

2 What is a neural network? [U] [1]

a) A set of algorithms to recognize patterns b) A type of biological brain

c) A hardware component in computers d) A collection of data structures

3 What is the main function of a biological neuron? [A] [1]

a) To transmit information b) To store information

c) To create memories d) To send electrical signals

4 What did Norbert Wiener develop in the 1940s? [K] [1]

a) Cybernetics b) The Turing Test

c) Machine Learning d) Deep Learning

5 Which AI system can improve over time by learning from data? [A] [1]

a) Supervised Learning b) Unsupervised Learning

c) Reinforcement Learning d) Deep Learning

6 Which of the following is an application of AI? [K] [1]

a) Social Media b) Mobile Apps

c) Weather Forecasting d) Messaging Apps

7 Which early AI program was developed for natural language processing? [U] [1]

a) MYCIN b) Logic Theorist

c) Perceptron d) ELIZA

8 Which is a key characteristic of AI systems? [U] [1]

a) Self-awareness b) Decision making

c) Problem-solving ability d) Learning from mistakes

9 What is the goal of natural language processing? [U] [1]

a) Mimic human emotions b) Understand human language

c) Recognize speech d) Translate human speech

10 What is the purpose of training a neural network? [U] [1]

a) To test its performance b) To initialize random weights

c) To minimize the loss function d) To increase network size

11 ChatGPT can assist in writing code in which programming language? [U] [2]

a) python b) javascript

c) c++ d) all of the above

12 Who developed ChatGPT? [K] [2]

a) Google b) Microsoft

c) OpenAI d) IBM

13 What is Playground AI primarily used for? [A] [2]

a) Text generation b) Image generation

c) Audio editing d) Video creation

14 What is DALL-E primarily used for? [K] [2]

a) Text editing b) Image generation

c) Video editing d) Data analysis

15 Which AI technique is commonly used in algorithmic trading? [K] [2]

a) Decision Trees b) Convolutional Neural Networks (CNNs)

c) Reinforcement Learning d) Support Vector Machines

16 Which AI application is used for analyzing text-based data like financial reports and news? [U] [2]

a) Support Vector Machines (SVM) b) Natural Language Processing (NLP)

c) Convolutional Neural Networks (CNNs) d) K-Means Clustering

17 How does AI assist with vehicle-to-everything (V2X) communication? [K] [2]

a) By allowing vehicles to communicate with infrastructure and other vehicles b) By increasing vehicle emissions

c) By replacing human drivers with robots d) By reducing the need for wireless networks

18 What AI technology helps detect anomalies in transportation systems for security purposes? [K] [2]

a) Anomaly detection algorithms b) Human inspection

c) Time series forecasting d) Blockchain technology

19 Which of the following is crucial for tool compatibility? [K] [2]

a) Ease of use b) Technical support

c) Integration with existing systems
d) Cost structure

20 What is the role of AI in video game development? [K] [2]
a) To automate game testing
b) To create non-player characters (NPCs) with realistic behavior
c) To generate dynamic content
d) All of the above

21 What type of AI model is DALL-E based on? [K] [2]
a) Generative Adversarial Network (GAN)
b) Convolutional Neural Network (CNN)
c) Transformer model
d) Recurrent Neural Network (RNN)

22 What command do you typically use to start generating an image in MidJourney? [A] [2]
a) /start
b) /generate
c) /imagine
d) /create

23 What type of language should users avoid when interacting with ChatGPT? [U] [2]
a) Professional language
b) Inappropriate or offensive language
c) Casual language
d) Technical jargon

24 What does POS tagging refer to in NLP? [U] [2]
a) Predicting the sentiment of a sentence
b) Parsing and splitting paragraphs
c) Identifying parts of speech in a sentence
d) Creating new words from a root word

25 What is Natural Language Generation (NLG) in content generation? [U] [2]
a) The process of generating human-like text from structured data
b) Translating text between languages
c) Creating visual designs
d) Analyzing the sentiment of text

26 What role does community support play in selecting an AI tool? [U] [2]
a) It enhances security
b) It provides quicker resolutions to issues
c) It increases the tool's popularity
d) It improves visual design

SERIES: A

27 In AI for healthcare, which of the following refers to the ethical issue of biased algorithms? [U] [2]

a) Patient confidentiality
b) Algorithmic transparency
c) Fairness in predictions
d) Data security

28 Which algorithm is used in supervised learning for classification? [U] [3]

a) Decision Tree
b) K-means
c) PCA
d) Random sampling

29 What is "training data"? [K] [3]

a) Data with known answers used to train the model
b) Data used to test the model
c) Data that the model generates
d) Data for cleaning the model

30 What is "exploitation"? [E] [3]

a) Trying something new
b) Using what is already known
c) Making mistakes
d) Ignoring rewards

31 What does fairness in AI mean? [E] [3]

a) Treating everyone the same, regardless of circumstances
b) Ensuring equal outcomes for everyone
c) Treating people in a way that is just and unbiased
d) Ignoring differences among people

32 What can happen if AI is not regulated? [E] [3]

a) Increased safety
b) Unfair practices and misuse
c) Better decision-making
d) More innovation

33 What is "responsible AI"? [U] [3]

a) AI that always makes the best choices
b) AI that considers ethical implications in decisions
c) AI that operates without any rules
d) AI that is difficult to understand

34 How can AI affect privacy? [K] [3]

a) By improving security
b) By collecting excessive personal data
c) By minimizing data usage
d) By enhancing user control

SERIES: A

35 What is "moral responsibility" in AI? [E] [3]

a) Ignoring the consequences of AI actions
b) Being accountable for the impact of AI
c) Focusing only on profits
d) Making decisions without considering ethics

36 What is the main challenge in Reinforcement Learning? [K] [3]

a) Finding the right data
b) Balancing trying new things and using known actions
c) Collecting enough rewards
d) Choosing the environment

37 How can AI create bias on social media? [E] [3]

a) By promoting diverse content
b) By favoring certain types of posts
c) By increasing user interaction
d) By filtering out all content

38 Which technique helps visualize data? [A] [3]

a) Clustering
b) Principal Component Analysis (PCA)
c) Regression
d) Classification

39 What is "targeted advertising"? [A] [3]

a) Ads for everyone
b) Ads aimed at specific user interests
c) Random ads shown to users
d) Ads based on location only

40 What does accountability mean in AI? [A] [3]

a) Blaming users for errors
b) Taking responsibility for AI actions
c) Ignoring ethical concerns
d) Increasing system complexity

(35 × 1 = 35)

END OF THE QUESTION PAPER

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Part A

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Answer any 35 questions. Each question carries one mark

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c) To initialize random weights d) To test its performance

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(CNNs) (SVM)

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c) By replacing human drivers d) By allowing vehicles to

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	c)	Regression	d)	Principal Component Analysis (PCA)	
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	a)	all of the above	b)	javascript	
	c)	c++	d)	python	
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